Summary

Research Objective

The main objective is to see if cross impact and integrated OFI can be used to generated trading signals and predict returns. The focus is on replication of core research findings [1]. The performance is assessed using cumulative returns.

Methods used

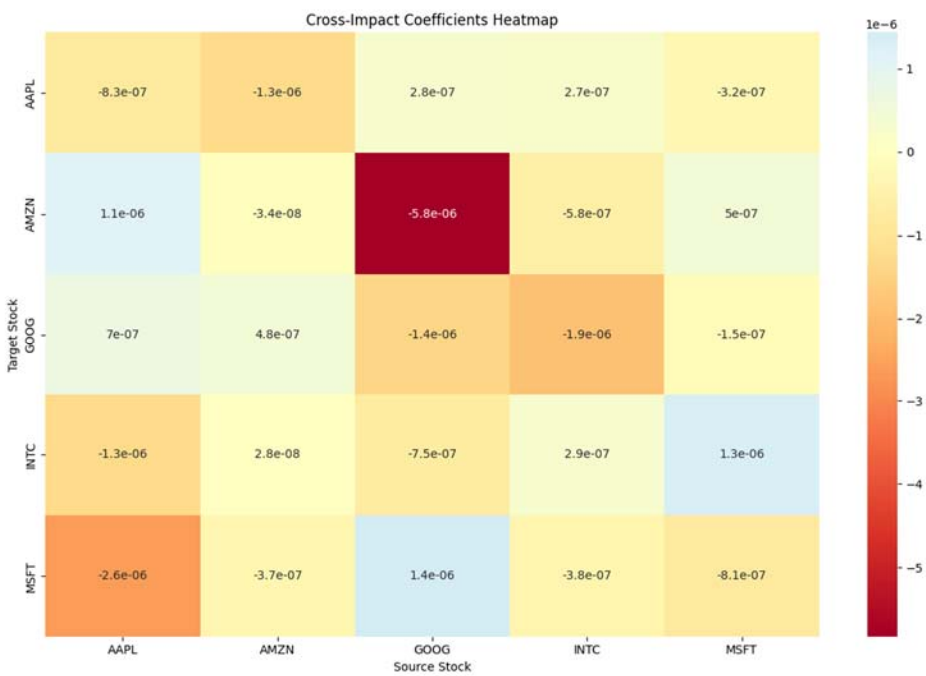
Stocks used: AAPL, AMZN, GOOG, INTC, MSFT

Data source: LOBSTER (with option to source from Yahoo Finance)

Models: Linear regression model

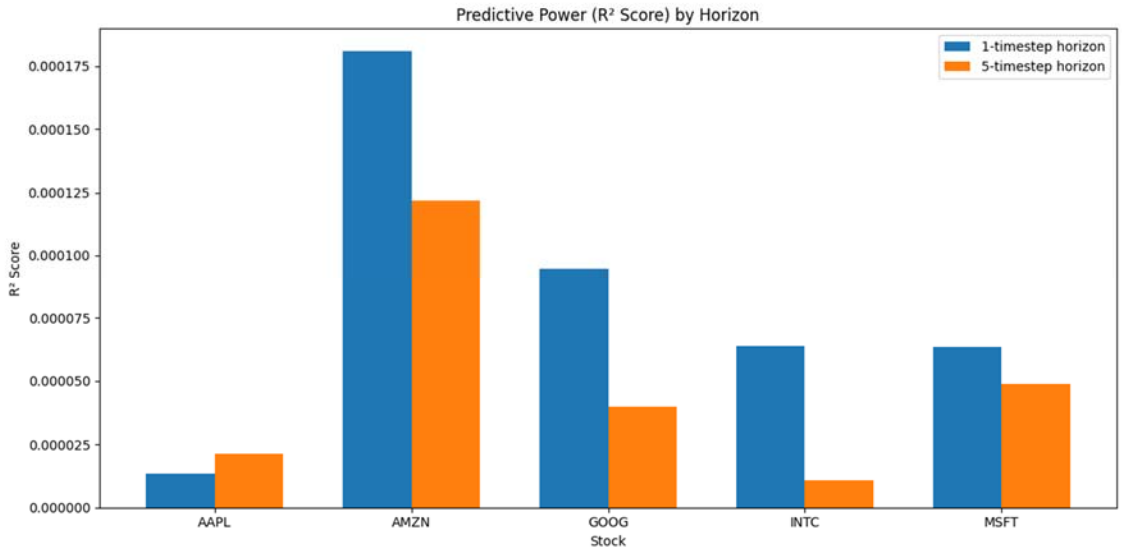
Strategy: Go long when signal exceeds the 70th percentile.

Analysis

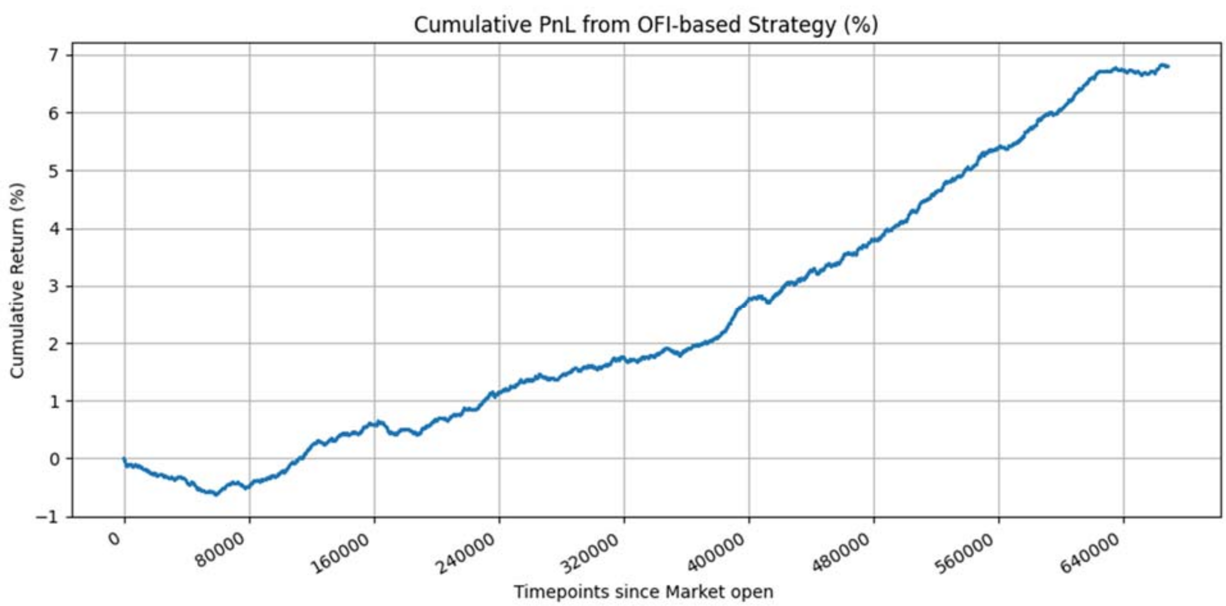


The impact of performance of one stock is most impactful on itself. Performance of one stock using OFI can not be used to measure impact on other stock across the board. There are minor impacts but these are also comparatively weak.

OFI tends to have a higher predictive power on 1 timestep ahead as opposed to 5 timesteps ahead as indicated by the research findings [1]. This highlights the importance have using OFI when executing quick and short-term trades.



Lastly using signals generated by OFI, algorithm trades are made and with a cumulative return of just under 7%.



Conclusion

Cross impact can be measure for a range of stock but the economic predictability is very low in most cases. But it could have some use cases as spoken about in the future works section.

OFI is very useful when trading on short term horizons with the predictive power degrading the longer the horizon. A trading strategy based on the top 30% of integrated OFI signals yields steady returns.

Future Work

- Gaining access to real time stock data as opposed to LOBSTER
- **Testing cross impact sector wise. Specifically, whether it is a useful metric assess a major event in the market in specific niches**
  If AAPL's returns are influenced by GOOG's OFI that suggests tech-sector events.
- Adding transactions cost to trades.
- Focusing on low volatility stocks as volatility adds noise to data

Code Availability

All source code for data ingestion, OFI computation, cross-impact analysis, and trading simulation is publicly available on GitHub: https://github.com/A-Binoy/CrossImpactAnalyzer

References

[1] R. Cont, M. Cucuringu, and C. Zhang, "Cross-impact of order flow imbalance in equity markets," *Quantitative Finance*, vol. 23, no. 10, pp. 1373–1393, 2023, doi: 10.1080/14697688.2023.2236159.